Four-Column Alignment: Functional Correspondence in Genomics as an Invariant of Network Response, Not of Sequence

Kundai Farai Sachikonye
kundai.sachikonye@wzw.tum.de

August 15, 2026

Abstract

Sequence comparison assumes that biological function is a property carried by a sequence and therefore recoverable from it. We argue that this assumption is not merely difficult to satisfy but structurally false, and we replace it with a criterion that does not require it. The argument begins from an elementary and universally accepted fact: every double-stranded nucleic acid consists of two strands that read as different codon sequences yet carry one message. If the message were a property of a codon string, the two strands would carry different messages. They do not. The message is therefore a property of the *relation* between the strands and not of either strand. We develop this observation into a general theory. Working entirely with finite weighted graphs, we prove that in a system whose parts are individuated by comparison against a complement that cannot be exhaustively enumerated, the cost of separating any part from the rest is bounded below by a strictly positive *floor* $\beta > 0$; that the resulting identity of a part is a minimum cut, invariant under relabelling, and never a point; and consequently that labels—sequences among them—carry no invariant. We then show that the quantity conventionally called the function of a sequence is not invariant either: it is receiver-relative, and distinct receivers register distinct values, each correct. What *is* invariant is the correspondence relation between two systems, and we give a procedure that certifies it without ever evaluating function. The procedure is a bidirectional relaxation over four columns: two comparing the units and two comparing the network responses those units induce. We prove that the relaxation is monotone, that it terminates or diverges with no third outcome, that its fixed point (*quiescence*) holds if and only if the two units induce the same response, and that the certificate is computed *blind*—no function is ever read. Four-column alignment separates two classes that sequence comparison cannot: near-identical sequences with divergent effect, and divergent sequences with identical effect. We give the algorithm, its complexity, the conditions under which it is well-founded, a falsifiable prediction, and an explicit account of what the framework does not establish. In particular we identify one assumption—response independence—on which the entire construction rests, show that it is empirically decidable, and specify the experiment that decides it.

Contents

Part I
The problem

1 Introduction

1.1 The question

Given two nucleic acid or protein sequences, the standard question is: how similar are they? A large and successful body of work answers it—global and local alignment [Needleman and Wunsch, 1970, Smith and Waterman, 1981], heuristic database search [Altschul et al., 1990, 1997], substitution matrices [Henikoff and Henikoff, 1992], and profile models [Eddy, 1998]. The answer is used as a proxy for a second question, which is the one actually of interest: do these two sequences *do the same thing*?

The proxy is known to be imperfect. It fails in two directions, and both failures are well documented:

(i) **Similar sequence, different function.** Synonymous substitutions—by construction invisible to the protein sequence—alter substrate specificity, folding kinetics, and expression level [Kimchi-Sarfaty et al., 2007, Sauna and Kimchi-Sarfaty, 2011, Plotkin and Kudla, 2011]. A comparison operating on sequence content cannot see the difference.

(ii) **Different sequence, same function.** Non-homologous isofunctional enzymes catalyse identical reactions with unrelated sequences and unrelated folds [Galperin et al., 1998, Omelchenko et al., 2010, Doolittle, 1994]. A comparison operating on sequence content cannot see the sameness.

These are usually treated as accuracy problems: the score is a noisy estimate of function, and better scores would estimate it better. We argue they are not accuracy problems. They are the visible consequence of a category error in the question, and no refinement of the score removes them.

1.2 The elementary observation

Consider a double-stranded nucleic acid [Watson and Crick, 1953]. Let the sequence of one strand be s and of its complement $\bar{s}$. Reading frames aside, s and $\bar{s}$ are different strings over the same alphabet; partitioned into codons they are different codon sequences. Yet the duplex carries one message, and no experiment distinguishes "the message of s" from "the message of $\bar{s}$" as two messages. Note also that s determines $\bar{s}$ and conversely, so neither strand is informationally privileged; Chargaff's regularities [Chargaff, 1950] are the population-level shadow of this.

Suppose the message were a property carried by a codon string. Then s and $\bar{s}$, being different codon strings, would carry different messages. They do not. Therefore the message is not a property carried by a codon string.

The inference is elementary, it uses no theory, and it is not controversial as a statement about molecules. What is not usually drawn is the consequence for methodology. If the message is not in either strand, then it is in what the two strands jointly instantiate—the complementarity relation. And a relation is not the kind of object that can be recovered from one relatum by examining it more closely. No amount of additional resolution applied to s alone will exhibit a property that s does not have.

Remark 1.1 (Scope of the observation). We do not claim that sequence is uninformative. It is highly informative: s constrains the relation sharply, which is why sequence comparison works as well as it does. We claim only that sequence is not *identical* to the object of interest, and that the residual gap between them is not a noise term that shrinks with better measurement.

1.3 What this paper does

We take the observation seriously and follow it to a working criterion. The development has four stages.

Stage 1 (Sections 2 to 4).
We formalise "individuation by comparison against a complement" in finite weighted graphs, and prove three results: a strictly positive floor $\beta > 0$ on separation cost; that identity is the minimum cut and is invariant under relabelling; and that labels carry no invariant. Nothing in this stage is biological.

Stage 2 (Section 5).
We show that function, as operationally defined by assay, is receiver-relative: it is a property of the pair (unit, network), not of the unit. We prove that no assignment of a single value to a unit is consistent with context-dependence, and that this is not remediable by more assays.

Stage 3 (Sections 6 and 7).
We construct the criterion. Two systems are compared not by content but by the *demand* each places on the other under bidirectional propagation. The construction is extended to four columns by adding the network responses the units induce. We prove monotonicity, a termination dichotomy, and that the fixed point certifies correspondence blind.

Stage 4 (Sections 8, 9 and 11).
We give the algorithm and its complexity, state a falsifiable prediction, and set out the assumptions—one of which is load-bearing and untested.

1.4 What this paper does not do

We state the negative claims at the outset, since a framework of this shape invites over-reading.

(1) We do not claim that function becomes computable from first principles. The construction requires a solved network model, which requires kinetic and thermodynamic parameters. We relocate the empirical burden from per-sequence assay to per-network parameterisation; we do not remove it. Section 11 is explicit.

(2) We do not claim to replace alignment. Alignment answers a question about sequence, and that question remains well posed and useful. We answer a different question.

(3) We do not claim the criterion is more accurate than curated annotation on any existing benchmark. No such comparison is performed here.

(4) We do not report a benchmark against biological ground truth. We do validate every theorem computationally, and we do run the decisive experiment of Section 9.1 on four solved networks—where it returns a *negative* result (Theorem 10.1). No claim of a working biological criterion is made.

Part II
Foundations

2 Contact graphs

We work throughout with finite weighted graphs [Diestel, 2017]. The constructions in this part are purely combinatorial and are stated without biological interpretation; the interpretation begins in Section 5.

Definition 2.1 (Finite weighted graph)**.** A *finite weighted graph* is a triple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, w)$ with $\mathcal{V}$ a finite non-empty set of *vertices*, $\mathcal{E} \subseteq \binom{\mathcal{V}}{2}$ a set of *edges*, and $w : \mathcal{E} \to \mathbb{R}_{>0}$ a strictly positive weight function. We write $n = |\mathcal{V}|$.

Strict positivity is the substantive hypothesis. It encodes the assumption that distinguishing two things is never free: an edge of weight zero would be a pair that can be told apart at no cost, which we exclude by fiat and justify in Section 3.

Definition 2.2 (Cut and cut weight)**.** For $S \subseteq \mathcal{V}$ the *cut* induced by S is

$$\partial(S) = \{\, \{u, v\} \in \mathcal{E} \ :\ u \in S,\ v \notin S \,\},$$

with *cut weight* $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. A cut *separates* u from v if $u \in S$ and $v \notin S$.

Definition 2.3 (Contact graph and medium)**.** A *contact graph* is a pair $(\mathcal{G}, \mathbf{m})$ where $\mathcal{G}$ is a finite weighted graph and $\mathbf{m} \in \mathcal{V}$ is a distinguished vertex, the *medium*, adjacent to every other vertex: $\{\mathbf{m}, v\} \in \mathcal{E}$ for all $v \in \mathcal{V} \setminus \{\mathbf{m}\}$. Vertices in $\mathcal{V} \setminus \{\mathbf{m}\}$ are called *items*.

The medium is the formal stand-in for "everything else". Its defining property is that every item is in contact with it: no item is isolated from the rest of the system.

Definition 2.4 (Separation cost)**.** The *separation cost* of an item v is the minimum weight of a cut separating v from the medium:

$$\sigma(v) \ = \ \min\{\, w(\partial(S)) \ :\ v \in S,\ \mathbf{m} \notin S \,\}.$$

A minimising set is written $S^*(v)$ and the corresponding cut $\partial(S^*(v))$ is a *minimum cut* of v.

By the max-flow min-cut theorem [Ford and Fulkerson, 1956] the quantity $\sigma(v)$ is computable in polynomial time; efficient algorithms are standard [Stoer and Wagner, 1997, Gomory and Hu, 1961]. We use computability only in Section 8; the theory does not depend on it.

Remark 2.5 (Why a graph)*.* The choice of finite weighted graphs is deliberate and restrictive. We avoid metric spaces (which would presuppose a distance we are trying to derive), probability spaces (which would presuppose a measure), and propositional logic (which would presuppose identity of terms). A graph with positive weights presupposes only that there are things, that some pairs can be told apart, and that telling apart has a cost.

3 The floor

We now derive the central structural fact. It is derived, not assumed: the positivity of edge weights in Theorem 2.1 is a modelling convention, but the *uniform* lower bound below follows from a hypothesis about enumeration.

3.1 Non-completability

Definition 3.1 (Expanding family; non-completable system)**.** An *expanding family* is a sequence of contact graphs $(\mathcal{G}_0, \mathbf{m}) \subseteq (\mathcal{G}_1, \mathbf{m}) \subseteq \cdots$ in which each $\mathcal{G}_{k+1}$ is obtained from $\mathcal{G}_k$ by adding vertices and edges, with weights of pre-existing edges unchanged. The family is *non-completable* if $|\mathcal{V}_k| \to \infty$.

The biological reading, stated once here and used in Section 5: a cell, an organism, or a sequence database is never a closed catalogue. New species, new interaction partners, new conditions are added indefinitely. No identification is ever performed against a finished inventory.

Definition 3.2 (Identification and its residual)**.** An *identification* of item v in $\mathcal{G}_k$ is the specification of a cut separating v from $\mathbf{m}$. Its *residual* is the weight of that cut. An identification is *exhaustive* if its residual is 0.

Theorem 3.3 (No exhaustive identification). *In a contact graph, no item admits an exhaustive identification.*

Proof. Let v be an item. By Theorem 2.3 the edge $\{\mathbf{m}, v\}$ exists, and every cut separating v from $\mathbf{m}$ contains at least one edge—in particular any cut $\partial(S)$ with $v \in S, \mathbf{m} \notin S$ contains the edge $\{\mathbf{m}, v\}$ if v's only neighbour in $\mathcal{V} \setminus S$ is $\mathbf{m}$, and in general contains at least the edges from S to $\mathcal{V} \setminus S$, a set that is non-empty because $\{\mathbf{m}, v\}$ joins S to its complement when $v \in S$ and $\mathbf{m} \notin S$. Since $w > 0$ on $\mathcal{E}$, the cut weight is strictly positive. Hence no cut has residual 0. □

Theorem 3.4 (Positive floor). *Let $(\mathcal{G}, \mathbf{m})$ be a contact graph. Then*

$$\beta \;:=\; \min_{e \in \mathcal{E}} w(e) \;>\; 0,$$

and for every item v one has $\sigma(v) \geq \beta$. We call β the contact floor.

Proof. $\mathcal{E}$ is finite and non-empty (it contains $\{\mathbf{m}, v\}$ for each item v; if there are no items the statement is vacuous), and w takes strictly positive values, so the minimum over a finite set of positive reals is attained and positive. For the second part, any cut separating v from $\mathbf{m}$ is a non-empty subset of $\mathcal{E}$ by the argument of Theorem 3.3, hence has weight at least $\min_{e \in \mathcal{E}} w(e) = \beta$. □

Corollary 3.5 (No sharp identification). *For every item v, $\sigma(v) \geq \beta > 0$. There is no identification of v of arbitrarily small cost, and in particular the identity of v cannot be a zero-content datum.*

Proof. Immediate from Theorem 3.4. □

3.2 Stability of the floor under expansion

Theorem 3.4 is a statement about a single graph. The following addresses what happens as the system grows, and it is the point at which we must be careful: a naive claim of invariance is false, and we state the true one.

Definition 3.6 (Local floor; intrinsic floor). For an item v in $\mathcal{G}_k$ put $\beta_k(v) = \min\{w(e) : e \in \mathcal{E}_k,\ v \in e\}$, the *local floor* of v at stage k. The *intrinsic floor* is $\beta^*(v) = \lim_{k \to \infty} \beta_k(v)$ when the limit exists.

Lemma 3.7 (Existence). *In any expanding family, $\beta^*(v)$ exists and satisfies $0 \leq \beta^*(v) \leq \beta_0(v)$.*

Proof. Adding edges incident to v can only lower the minimum incident weight, and existing weights are unchanged, so $(\beta_k(v))_k$ is non-increasing. It is bounded below by 0. A non-increasing sequence bounded below converges. □

Remark 3.8 (The intrinsic floor need not be positive). It is tempting to assert $\beta^*(v) \geq \beta_0 > 0$, but this is false in general. If the expansion introduces edges at v of weight $1/k$ at stage k, then $\beta_k(v) \to 0$ and $\beta^*(v) = 0$: the item *dissolves* into the medium in the limit, becoming indistinguishable from it at arbitrarily small cost. What is true, and all that we use, is that $\beta_k(v) \geq \beta(\mathcal{G}_k) > 0$ at every finite stage (Theorem 3.4). Individuation holds at every finite stage regardless of whether it survives the limit.

Definition 3.9 (Floor-stabilising family). An expanding family is *floor-stabilising at v* if there exists K with $\beta_k(v) = \beta_K(v)$ for all $k \geq K$; equivalently, if no expansion after stage K introduces an edge at v of weight below $\beta_K(v)$.

Theorem 3.10 (Invariance under expansion). *Let an expanding family be floor-stabilising at v, with stabilisation index K. Then:*

(i) $\beta^*(v) = \beta_K(v) > 0$;

(ii) *$\beta^*(v)$ is independent of $|\mathcal{V}_k|$ for $k \geq K$: it takes the same value in every subsequent enlargement of the system;*

(iii) *every other quantity attached to v—its neighbourhood $N_k(v)$, its separation cost $\sigma_k(v)$, its minimum cut $\partial(S_k^*(v))$—may change at every stage $k \geq K$.*

Proof. (i) By Theorem 3.9 the sequence is eventually constant at $\beta_K(v)$, which is positive by Theorem 3.4 applied to $\mathcal{G}_K$. (ii) Constancy of the sequence for $k \geq K$ is exactly independence of the stage index, hence of $|\mathcal{V}_k|$ along the family. (iii) It suffices to exhibit changes. $N_k(v)$ grows whenever an edge at v is added, which the definition of an expanding family permits. For $\sigma_k(v)$: adding a new item u adjacent to both v and $\mathbf{m}$ with $\{v, u\}$ of large weight increases the weight of every cut placing u outside S, and adding u adjacent to $\mathbf{m}$ only, with a light edge to v, can create a cheaper cut placing u inside $S^*(v)$; in either case the minimising set and its weight may change while $\beta_k(v)$ is untouched. □

Remark 3.11 (What Theorem 3.10 claims). The theorem is conditional on floor-stabilisation, which is a hypothesis about the expansion and not a theorem about graphs. We flag this because the unconditional statement is false (Theorem 3.8). Section 11 records the biological reading of the hypothesis: it asserts that the cost of distinguishing a molecular species from its nearest confusable neighbour does not fall to zero as the catalogue of known species grows. This is an empirical claim about chemistry, and it can fail—two species may prove to be indistinguishable under all available observables—in which case the framework correctly reports dissolution rather than identity.

4 Identity, and why labels carry none

Definition 4.1 (Weighted isomorphism; relabelling)**.** A *weighted isomorphism* $\varphi : (\mathcal{G}, \mathbf{m}) \to (\mathcal{G}', \mathbf{m}')$ is a bijection $\varphi : \mathcal{V} \to \mathcal{V}'$ with $\varphi(\mathbf{m}) = \mathbf{m}'$, $\{u, v\} \in \mathcal{E} \iff \{\varphi u, \varphi v\} \in \mathcal{E}'$, and $w'(\{\varphi u, \varphi v\}) = w(\{u, v\})$. A *relabelling* is a weighted isomorphism from $(\mathcal{G}, \mathbf{m})$ to itself.

Theorem 4.2 (The minimum cut is the invariant; the label is not)**.** *Let $\varphi : (\mathcal{G}, \mathbf{m}) \to (\mathcal{G}', \mathbf{m}')$ be a weighted isomorphism. Then for every item v,*

$$\sigma_{\mathcal{G}'}(\varphi v) = \sigma_{\mathcal{G}}(v), \qquad \varphi(\partial(S^*(v))) = \partial(S^*(\varphi v)).$$

By contrast, the vertex label of v is not preserved by relabellings: any non-trivial automorphism permutes labels while fixing all cut weights.

Proof. φ maps cuts to cuts bijectively and preserves weights, so it maps the family of v–$\mathbf{m}$ cuts onto the family of φv–$\mathbf{m}'$ cuts weight-for-weight; minima therefore agree, and minimisers correspond. For the second statement, let φ be a non-trivial automorphism; then $\varphi v \neq v$ for some v, so the label changes while, by the first part, every cut weight is unchanged. □

Theorem 4.3 (Identity is a region, never a point)**.** *The invariant data fixing an item v is the minimum cut $\partial(S^*(v))$, a set of edges of total weight $\sigma(v) \geq \beta > 0$. It is in general not carried by any single vertex: there exist contact graphs in which $S^*(v) \neq \{v\}$ for every minimiser.*

Proof. Invariance is Theorem 4.2; positivity is Theorem 3.4. For non-singleton minimisers, take two clusters C_1, C_2 of r items each, all intra-cluster edges of weight W, a single edge between the clusters of weight β, and edges from $\mathbf{m}$ to each item of weight β. For $v \in C_1$, the cut $\partial(\{v\})$ has weight $\beta + (r-1)W$ (its edge to the medium plus its intra-cluster edges), whereas $\partial(C_1)$ has weight $r\beta + \beta$ (medium edges plus the inter-cluster edge). For $W > \beta(r+1)/(r-1)$ the latter is strictly smaller, so every minimiser contains all of C_1 and $S^*(v) \neq \{v\}$. □

Corollary 4.4 (Labels carry no invariant)**.** *No function of the vertex label alone is invariant under relabelling. In particular, if two systems are related by a weighted isomorphism, any quantity computed from labels may differ between them while every invariant agrees.*

Proof. A non-trivial automorphism fixes all invariants (Theorem 4.2) and moves labels, so a label-function taking distinct values on v and φv is not invariant. $\square$

Remark 4.5 (The bearing on sequence). Theorem 4.4 is the abstract form of the observation of Section 1. A sequence is a label: a name for an item, drawn from an alphabet, which the system's structure does not determine. The two strands of a duplex are two labels for one relation, related by the complementation map, which is precisely a relabelling in the sense of Theorem 4.1 when lifted to the contact graph on molecular species. That the labels differ while the message does not is exactly Theorem 4.4.

Part III
Function

5 Function is receiver-relative

We now connect the combinatorics to biology. The connection is made through a single modelling assumption, stated explicitly so that it can be challenged.

Assumption 5.1 (Network representation)**.** A biological context—a cell type, a tissue, a condition—is represented by a *network* $\mathcal{N} = (\mathcal{G}, \mathbf{m})$: a contact graph whose items are molecular species and whose edge weights $w(\{u, v\})$ measure the minimum cost of distinguishing species u from species v within that context. A *unit* (a sequence, a variant, a locus) acts on $\mathcal{N}$ by inducing a perturbation.

The edge weights are physical, not conventional: a natural realisation takes w from thermodynamic separations—differences of chemical potential, conductances derived from kinetic constants—so that the contact graph of a metabolic network is a function of the solved steady state [Beard et al., 2002, Qian and Beard, 2005, Orth et al., 2010], with parameters available from public resources [Kanehisa and Goto, 2000, Jassal et al., 2020, Schomburg et al., 2002]. We use no property of the realisation beyond Theorem 5.1.

Definition 5.2 (Perturbation; response)**.** A *perturbation* of $\mathcal{N}$ is a map $\Delta : \mathcal{E} \to \mathbb{R}_{\geq 0}$, acting by $w \mapsto w + \Delta$. The *response* of $\mathcal{N}$ to a unit x is the perturbation Δ_x that x induces. A response is *non-trivial* if $\Delta_x \not\equiv 0$.

Perturbations are non-negative: they raise the cost of distinctions. This is the graph-level statement of a familiar thermodynamic fact—a physical operation may blur a distinction but cannot create separation for free [Landauer, 1961, Bennett, 1982, Parrondo et al., 2015]. We note the consequence that concerns us and no more.

Proposition 5.3 (The skeleton persists; the weights move)**.** *Let Δ be a perturbation of $\mathcal{N}$. Then $\mathcal{E}$ is unchanged and $\beta(\mathcal{N})$ is non-decreasing, while $\sigma(v)$ may change for every item v.*

Proof. Δ alters w and not $\mathcal{E}$, so the edge set is unchanged, and since $\Delta \geq 0$ every edge weight is non-decreasing, hence so is their minimum. For the last clause, raising the weight of any edge in $\partial(S^*(v))$ raises $\sigma(v)$ unless a different cut becomes minimal, and either way the value may change. $\square$

Remark 5.4. Theorem 5.3 is the formal content of a claim that is intuitive but rarely stated precisely: a perturbed system is not a *different* system but the *same* system with certain distinctions made more expensive. Which distinctions have become expensive is the informative datum.

5.1 Receivers

Definition 5.5 (Receiver; registered value)**.** A *receiver* is a network $\mathcal{N}$ together with a readout $\rho_{\mathcal{N}}$ mapping perturbations to observations. The *registered value* of a unit x at receiver $\mathcal{N}$ is $\rho_{\mathcal{N}}(\Delta_x^{\mathcal{N}})$, i.e. the observable consequence of x's response in that network.

Assays are receivers. A transcriptomic readout in one tissue and the same readout in another are two receivers, because the networks differ.

Theorem 5.6 (No privileged value)**.** *Let x be a unit and let $\mathcal{N}_1, \mathcal{N}_2$ be receivers with $\Delta_x^{\mathcal{N}_1} \neq \Delta_x^{\mathcal{N}_2}$ non-trivial. Then in general $\rho_{\mathcal{N}_1}(\Delta_x^{\mathcal{N}_1}) \neq \rho_{\mathcal{N}_2}(\Delta_x^{\mathcal{N}_2})$, and there is no assignment $x \mapsto \phi(x)$ of a single value to units such that $\rho_{\mathcal{N}}(\Delta_x^{\mathcal{N}}) = \phi(x)$ for all receivers $\mathcal{N}$, unless the response of x is the same in every network.*

Proof. Suppose such a ϕ existed. Then for all $\mathcal{N}_1, \mathcal{N}_2$ we would have $\rho_{\mathcal{N}_1}(\Delta_x^{\mathcal{N}_1}) = \phi(x) = \rho_{\mathcal{N}_2}(\Delta_x^{\mathcal{N}_2})$, i.e. the registered value is independent of the network. Since $\rho_{\mathcal{N}}$ is by Theorem 5.5 a function of the response *in* $\mathcal{N}$, and the responses differ by hypothesis, constancy of the composite for all choices of ρ forces $\Delta_x^{\mathcal{N}_1} = \Delta_x^{\mathcal{N}_2}$, contrary to hypothesis. Hence no such ϕ exists. □

Corollary 5.7 (Context-dependence is not error)**.** *Divergent measurements of "the function of x" across tissues or conditions are not measurement error and are not resolved by more or better assays. They are distinct registered values at distinct receivers, each correct for its receiver.*

Proof. Immediate from Theorem 5.6: the values are values of different functions, so their disagreement carries no implication of error. □

This is the theoretical counterpart of a large body of observation: tissue-specific effects of the same variant [GTEx Consortium, 2020], incomplete penetrance [Cooper et al., 2013], and pervasive genetic interaction [Lehner, 2011, Domingo et al., 2019, Costanzo et al., 2016]. The framework predicts these rather than accommodating them.

5.2 The circularity of annotation by assay

Proposition 5.8 (Annotation presupposes completability)**.** *Let x be a unit and suppose its function is defined as the registered value at some receiver $\mathcal{N}$. Then the assignment of that value to x as a property of x is licensed only if the response of x is invariant across all receivers in the expanding family—that is, only if the family of contexts is effectively complete.*

Proof. By Theorem 5.6 a single value is consistent with all receivers only when the response is network-independent. Absent that, the value attaches to the pair $(x, \mathcal{N})$. Asserting it of x alone is therefore an implicit quantification over all $\mathcal{N}$, which in a non-completable family (Theorem 3.1) cannot be discharged by finitely many observations. □

Corollary 5.9 (The two failure modes)**.** *The failure modes of Section 1(i) and (ii) are consequences of Theorem 4.4 and Theorem 5.6 respectively: (i) compares labels and therefore misses a difference that lives in the response; (ii) compares labels and therefore misses a sameness that lives in the response.*

Remark 5.10 (On missing heritability). We note without developing it that Theorem 5.6 bears on the observation that per-variant effects sum to a fraction of trait variance [Manolio et al., 2009, Boyle et al., 2017]: if the effect attaches to a (unit, network) pair, then a sum of per-unit effect estimates is a sum over objects that do not carry the quantity being summed. We do not claim this resolves the discrepancy, and we perform no analysis of it here.

Part IV
The criterion

6 Alignment as demand, not as content

If function cannot be read off a unit, comparison of functions cannot proceed by reading and comparing them. We therefore construct a comparison that never evaluates function.

6.1 Alignment and distance to the floor

Definition 6.1 (Alignment)**.** Fix a network $\mathcal{N}$ with floor β and let $\Omega = \sum_{e\in\mathcal{E}} w(e)$. For items u, v the *alignment* is

$$a(u,v) \;=\; \frac{\sigma_{\mathcal{N}}(u \mid v)}{\Omega} \in \left[\tfrac{\beta}{\Omega}, 1\right],$$

where $\sigma_{\mathcal{N}}(u \mid v)$ is the minimum weight of a cut separating u from v. The *distance to the floor* is $a(u,v) - \beta/\Omega \geq 0$.

Alignment is a normalised separation, not a similarity of content: it measures what it costs to tell u from v in the network, and it is bounded below by the floor and never zero (Theorem 3.4). It requires no probability model, no substitution matrix, and no free parameter.

Proposition 6.2 (Distance to the floor is parameter-free and vanishes only at coincidence)**.** *$a(u,v) - \beta/\Omega = 0$ if and only if u and v are separated at exactly the floor, i.e. the cheapest available distinction. The quantity is determined by $(\mathcal{G}, w)$ alone.*

Proof. Both claims are immediate from Theorem 6.1 and Theorem 3.4: the expression is a difference of two quantities each computed from the weighted graph, and it vanishes exactly when $\sigma_{\mathcal{N}}(u \mid v) = \beta$. □

6.2 Columns, anchors, and residue

Definition 6.3 (Column)**.** A *column* is a pair $(x, \mathcal{N})$ of a unit and a network, together with a partition of the items of $\mathcal{N}$ into *anchors* $\mathrm{anc}(x)$ and *residue* $\mathrm{res}(x)$. The anchors are the items whose removal most reduces $\sigma(x)$; formally, fix $r \in \mathbb{N}$ and let $\mathrm{anc}(x)$ be the r items u maximising the ablation score $\sigma_{\mathcal{N}}(x) - \sigma_{\mathcal{N}\setminus u}(x)$, ties broken by a fixed order. The residue is everything else.

The split is the formal counterpart of a familiar move: hold fixed the entities, and compare the relations among them. Anchors are the entities whose identity is not in question; residue is the relational structure the comparison must reconcile.

Definition 6.4 (Cross-demand)**.** Let $\mathsf{A} = (x_1, \mathcal{N}_1)$ and $\mathsf{B} = (x_2, \mathcal{N}_2)$ be columns and let $\pi_{\mathsf{A}\to\mathsf{B}}$ be a map carrying residue items of A to items of $\mathcal{N}_2$, holding anchors fixed (a *correspondence*). The *cross-demand* from A to B is

$$\mathrm{d}_{\mathsf{A}\to\mathsf{B}} \;=\; \sum_{i\in\mathrm{res}(\mathsf{B})} \Big(a\big(i, \pi_{\mathsf{A}\to\mathsf{B}}(i)\big) - \tfrac{\beta}{\Omega}\Big) \;\geq\; 0. \tag{1}$$

Column B is *satisfied from* A when $\mathrm{d}_{\mathsf{A}\to\mathsf{B}} = 0$.

Cross-demand is the total above-floor cost of making B consistent with A's image. It is zero exactly when every residue item of B sits at the floor relative to its correspondent—that is, when nothing further can be demanded.

6.3 The relaxation

Definition 6.5 (Relaxation; quiescence)**.** The *relaxation* of a pair (A, B) alternates two updates: B's residue is revised to reduce $\mathrm{d}_{\mathsf{A}\to\mathsf{B}}$, then re-ablated to recompute anchors; then symmetrically for A. The pair is *quiescent* when $\mathrm{d}_{\mathsf{A}\to\mathsf{B}} = \mathrm{d}_{\mathsf{B}\to\mathsf{A}} = 0$. A correspondence is *proper* iff the relaxation reaches quiescence.

To obtain termination we require that updates not idle. The following is the minimal such condition and we adopt it explicitly, since without it the dichotomy below is false.

Assumption 6.6 (Effective update)**.** Every update applied at a non-quiescent state strictly decreases the total demand $D = \mathrm{d}_{\mathsf{A}\to\mathsf{B}} + \mathrm{d}_{\mathsf{B}\to\mathsf{A}}$ by at least $\eta > 0$, where η is fixed in advance.

Theorem 6.7 (Monotonicity)**.** *Under Theorem 6.6, the sequence of total demands $D_0, D_1, D_2, \ldots$ generated by the relaxation is strictly decreasing until quiescence, and is bounded below by* 0*.*

Proof. Each $D_t \geq 0$ by Theorem 6.4, since it is a sum of non-negative terms. If the state at step t is not quiescent then $D_t > 0$ and by Theorem 6.6 $D_{t+1} \leq D_t - \eta < D_t$. ☐

Theorem 6.8 (Termination dichotomy)**.** *Under Theorem 6.6 the relaxation has exactly two possible outcomes:*

(i) *it reaches quiescence after at most $\lceil D_0/\eta \rceil$ updates; or*

(ii) *it halts at a non-quiescent state from which no effective update exists.*

In particular the relaxation cannot cycle indefinitely among non-quiescent states.

Proof. Suppose the relaxation performs T updates without reaching quiescence. By Theorem 6.7, $D_T \leq D_0 - T\eta$, and $D_T \geq 0$, so $T \leq D_0/\eta$. Hence at most $\lceil D_0/\eta \rceil$ effective updates occur. After the last one, either the state is quiescent—case (i)—or it is not and no effective update is available, since any such update would extend the sequence, contradicting maximality—case (ii). Cycling is excluded because D is strictly decreasing along updates and therefore never revisits a value. ☐

Remark 6.9 (Case (ii) is a first-class outcome). Case (ii) is not a failure of the procedure but an informative verdict: the two systems admit no correspondence reconcilable by the available moves. We call this an *honest decline*. A criterion that must emit a score for every input cannot express it, and is obliged to return a low-confidence number where the correct answer is a refusal. We return to this in Section 9.

Theorem 6.10 (Quiescence certifies blind)**.** *Let (A, B) be quiescent. Then for every residue item of either column the alignment to its correspondent sits at the floor. Conversely, if some residue item is separated from its correspondent above the floor, the pair is not quiescent. Moreover the certificate is computed without evaluating $\rho_{\mathcal{N}}$ for either column: only cut weights enter* (1)*.*

Proof. Quiescence is $\mathrm{d}_{\mathsf{A}\to\mathsf{B}} = \mathrm{d}_{\mathsf{B}\to\mathsf{A}} = 0$. Each demand is a sum of non-negative terms by Theorem 6.4, so it vanishes iff every term vanishes, i.e. iff $a(i, \pi(i)) = \beta/\Omega$ for every residue item—the floor. The converse is the contrapositive: a term exceeding β/Ω makes the sum positive. For blindness, inspect (1): it is built from a, which by Theorem 6.1 is a normalised minimum cut weight, and from β and Ω, all functions of $(\mathcal{G}, w)$; the readout $\rho_{\mathcal{N}}$ of Theorem 5.5 does not appear. ☐

Theorem 6.11 (Form survives: quiescence at positive residual)**.** *At quiescence the total separation between the two columns is at least $\beta \cdot |\mathrm{res}| > 0$. Quiescence is therefore never coincidence of content: the columns remain distinguishable while placing no demand on each other.*

Proof. By Theorem 3.4 every maintained separation carries weight at least β; there are $|\mathrm{res}|$ residue correspondences, each contributing at least β to the total separation. Positivity follows from $\beta > 0$. That this is compatible with zero demand is Theorem 6.10: demand measures the *above-floor* excess, which vanishes while the floor term remains. □

Theorem 6.11 is what distinguishes this criterion from a similarity score. Two systems can place no demand on one another—agree completely in the relevant sense—while remaining irreducibly distinct in form. A score that reached its extremum only at identity of content could not express this.

7 Four-column alignment

Theorem 6.10 certifies that two units correspond as *units*. It does not yet certify that they *do the same thing*, because two units may correspond in structure and diverge in what they induce. We therefore add the responses.

Construction 7.1 (The four columns)**.** Given units x_1, x_2 in networks $\mathcal{N}_1, \mathcal{N}_2$, form:

- **central columns** $\mathsf{A} = (x_1, \mathcal{N}_1)$ and $\mathsf{B} = (x_2, \mathcal{N}_2)$;
- **response columns** $\mathrm{R}(\mathsf{A}) = (\Delta_{x_1}^{\mathcal{N}_1}, \mathcal{N}_1)$ and $\mathrm{R}(\mathsf{B}) = (\Delta_{x_2}^{\mathcal{N}_2}, \mathcal{N}_2)$, the perturbations the units induce, treated as columns via Theorem 6.3 applied to the perturbed networks.

The relaxation is run over all four columns, with cross-demands computed for the central pair and for the response pair.

Definition 7.2 (Four-column quiescence)**.** The system is *four-column quiescent* if both (A, B) and $(\mathrm{R}(\mathsf{A}), \mathrm{R}(\mathsf{B}))$ are quiescent.

Theorem 7.3 (The four-column criterion)**.** *The system is four-column quiescent if and only if* (a) *the units correspond, and* (b) *the responses they induce correspond. Both conjuncts are certified blind, and (b) is not implied by (a).*

Proof. The equivalence with (a) and (b) is Theorem 7.2 together with Theorem 6.10 applied to each pair; blindness is the last clause of Theorem 6.10. For the non-implication, it suffices to exhibit a case with (a) and not (b), which is Theorem 7.4 below. □

Theorem 7.4 (Detection of aligned-content, divergent-response pairs)**.** *There exist units x_1, x_2 whose central columns are quiescent while their response columns are not. For such pairs, any criterion depending only on the central comparison reports correspondence, and the four-column criterion reports its absence.*

Proof. Construct $\mathcal{N}_1 = \mathcal{N}_2 = \mathcal{N}$ on items $\{u_1, u_2, z, w\}$ with

$$w(\{\mathbf{m}, u_1\}) = w(\{\mathbf{m}, u_2\}) = \beta, \qquad w(\{\mathbf{m}, z\}) = w(\{z, w\}) = W > \beta,$$

and no other edges. Because u_1 and u_2 are pendant on the medium, the cheapest cut separating them isolates one of them at its single incident edge, so $\sigma_{\mathcal{N}}(u_1 \mid u_2) = \beta$ exactly; by Theorem 6.10 the central columns for $x_1 \mapsto u_1$, $x_2 \mapsto u_2$ have zero demand. Now let x_1 induce Δ_1 supported on $\{z, w\}$ with $\Delta_1(\{z, w\}) = c > 0$, and let x_2 induce $\Delta_2 \equiv 0$. The pair (z, w) is separated above the floor in both response networks, and raising $w(\{z, w\})$ by c strictly increases $a(z, w)$ while the floor is pinned at β by the pendant edges; hence the two response columns register different above-floor gaps and the response cross-demand is positive. The central comparison is untouched by Δ_1, which is supported away from u_1, u_2. □

Remark 7.5 (A witness that does *not* work). It is tempting to join u_1 and u_2 by an edge of weight β and conclude that they are "separated at the floor". They are not: the minimum cut is free to ignore that edge and isolate u_1 through its other incident edges, which in general costs strictly more than β, leaving the central demand positive. The separation cost is a property of the cheapest *cut*, not of any single edge—an instance of Theorem 4.3. The pendant construction above avoids this because there the cheapest cut and the single incident edge coincide. This distinction was found by the validation suite of Section 10 and is recorded here because it is exactly the error the region-valued view of identity is meant to prevent.

Corollary 7.6 (The converse case)**.** *Symmetrically there exist units whose central columns are not quiescent while their response columns are: units of divergent content inducing corresponding responses. Sequence-content comparison reports no correspondence; the response columns report one.*

Proof. Take items $\{p, u_1, u_2, q, z, w\}$ with $w(\{\mathbf{m}, p\}) = \beta$—so that p realises the floor—and $w(\{\mathbf{m}, u_i\}) = w(\{u_i, q\}) = w(\{\mathbf{m}, q\}) = W$ for $i = 1, 2$ with $W > \beta$. Isolating either u_i now costs $2W > \beta$, so $\sigma(u_1 \mid u_2) > \beta$ and the central demand is strictly positive by Theorem 6.10. Let both units induce the same perturbation on $\{z, w\}$; the two response networks are then equal, so every response cross-demand computed against a fixed correspondence agrees, and the response pair is quiescent. □

Theorem 7.4 and Theorem 7.6 are the two failure modes of Section 1, now separated by construction rather than approximated. This is the substantive claim of the paper: the two classes that sequence comparison provably conflates are exactly the two that the response columns distinguish.

Remark 7.7 (Why the response and not the outcome). The criterion compares what the units *demand of their networks*, not what the networks are observed to do. The distinction matters because the observed outcome is a registered value and therefore receiver-relative (Theorem 5.6), whereas the demand is a cut computation in the network and is invariant under relabelling (Theorem 4.2). Comparing demands is comparing invariants; comparing outcomes is comparing labels.

7.1 The load-bearing assumption

Assumption 7.8 (Response independence)**.** For a unit x and network $\mathcal{N}$, any two admissible constructions of the response $\Delta_x^{\mathcal{N}}$ yield the same cross-demand against a fixed comparison column, up to a tolerance below the decision threshold.

Proposition 7.9 (What depends on Theorem 7.8)**.** *If Theorem 7.8 fails, then the verdict of Theorem 7.3 depends on which response was constructed, and the criterion is not well defined as a function of* $(x_1, \mathcal{N}_1, x_2, \mathcal{N}_2)$.

Proof. Suppose two admissible responses Δ, Δ' for the same $(x, \mathcal{N})$ give cross-demands differing by more than the decision threshold against a fixed column. Then the response pair is quiescent under one and not the other, so by Theorem 7.2 the four-column verdict differs while the inputs are unchanged. □

Theorem 7.8 is the single assumption on which the criterion's well-definedness rests. It is not a theorem of this paper, and we do not assert it. It is empirically decidable, and Section 9 specifies the experiment.

Algorithm 1 Four-column alignment.

Require: Units x_1, x_2; networks $\mathcal{N}_1, \mathcal{N}_2$; anchor count r; threshold θ; step floor η
Ensure: CORRESPOND, DIVERGE, or DECLINE

$\mathsf{A} \leftarrow \textsc{Column}(x_1, \mathcal{N}_1, r)$; $\quad \mathsf{B} \leftarrow \textsc{Column}(x_2, \mathcal{N}_2, r)$
$\mathrm{R}_1 \leftarrow \textsc{Column}(\Delta_{x_1}^{\mathcal{N}_1}, \mathcal{N}_1, r)$; $\quad \mathrm{R}_2 \leftarrow \textsc{Column}(\Delta_{x_2}^{\mathcal{N}_2}, \mathcal{N}_2, r)$
$D \leftarrow \mathrm{d}_{\mathsf{A}\to\mathsf{B}} + \mathrm{d}_{\mathsf{B}\to\mathsf{A}} + \mathrm{d}_{\mathrm{R}_1\to\mathrm{R}_2} + \mathrm{d}_{\mathrm{R}_2\to\mathrm{R}_1}$
while $D > \theta$ **do**
    apply an effective update (Theorem 6.6) to the pair with larger demand
    recompute anchors by ablation; recompute D
    if no effective update exists **then**
        return DECLINE $\qquad \triangleright$ Theorem 6.8(ii)
    end if
end while
if central and response demands both $\leq \theta$ **then**
    return CORRESPOND
else
    return DIVERGE
end if

Part V
Algorithm, predictions, limits

8 Algorithm and complexity

Proposition 8.1 (Complexity)**.** *Let* $n = \max(|\mathcal{V}_1|, |\mathcal{V}_2|)$ *and let* $M(n)$ *be the cost of one minimum-cut computation. Algorithm 1 performs* $\mathcal{O}(rn)$ *cut computations per ablation round and terminates after at most* $\lceil D_0/\eta \rceil$ *updates, giving total cost* $\mathcal{O}(rn\, M(n)\, D_0/\eta)$*.*

Proof. Anchor selection evaluates the ablation score for each of n candidate items, each requiring one separation-cost computation, and retains the top r; this is $\mathcal{O}(n)$ cut computations per round, and the r retained anchors are re-checked, giving $\mathcal{O}(rn)$ in the worst case. The update count is Theorem 6.8(i). Multiplying gives the stated bound. □

With $M(n)$ polynomial [Stoer and Wagner, 1997], the procedure is polynomial. We make no claim of practicality at genome scale; the bound is stated to show the criterion is effective, not to advertise performance.

Remark 8.2 (Direction of use). The criterion is naturally run *backward*: given a response, find the units that could have induced it. This is the direction in which the anchor/residue split is informative, since the anchors are read off the response and the residue is what must be reconciled. Forward construction—computing a response from a unit—requires solving the network and is correspondingly more expensive.

9 Predictions and the decisive experiment

We state what the framework forbids, since that is what makes it testable.

P1. Synonymous divergence is detectable. There exist variant pairs, identical at the protein-sequence level, for which the response columns are non-quiescent. Sequence-based criteria classify them as equivalent; the four-column criterion does not. Candidate instances are documented [Kimchi-Sarfaty et al., 2007, Sauna and Kimchi-Sarfaty, 2011].

P2. Non-homologous equivalence is detectable. There exist sequence pairs below any conventional homology threshold whose response columns are quiescent. Instances are documented [Galperin et al., 1998, Omelchenko et al., 2010].

P3. Verdicts are context-scoped. The same unit pair may be CORRESPOND in one network and DIVERGE in another. By Theorem 5.6 this is predicted, not anomalous; a framework claiming context-free verdicts is thereby distinguished from this one.

P4. A non-empty decline class. Some unit pairs yield DECLINE (Theorem 6.8(ii)). The framework predicts this class is non-empty and that it overlaps substantially with variants currently classified as of uncertain significance [Richards et al., 2015, Landrum et al., 2018, Hoffman-Andrews, 2017]. If every pair resolved, the dichotomy would be vacuous and the framework weakened.

Remark 9.1 (Falsification). P1 and P2 are refuted by exhibiting the documented instances and finding the response columns behave oppositely to the prediction. P4 is refuted by a decline class that is empty or that fails to enrich for genuinely ambiguous cases.

9.1 The experiment that decides Theorem 7.8

Everything above is conditional on response independence (Theorem 7.9). We specify its test.

Construction 9.2 (Response-independence test)**.** Fix a solved network $\mathcal{N}$ and an item x. Generate two admissible responses Δ, Δ' for x by distinct routes—for instance by perturbing distinct edges incident to x that a mechanistic model regards as equally valid realisations of the same intervention. Fix a comparison column B and compute $\mathrm{d}_{\Delta\to\mathsf{B}}$ and $\mathrm{d}_{\Delta'\to\mathsf{B}}$. Repeat across items and networks and report the distribution of $|\mathrm{d}_{\Delta\to\mathsf{B}} - \mathrm{d}_{\Delta'\to\mathsf{B}}|$ relative to the decision threshold θ.

Proposition 9.3 (Decisiveness)**.** *If the discrepancy distribution of Theorem 9.2 lies below θ, Theorem 7.8 holds at that threshold and Theorem 7.3 is well defined. If it does not, the criterion requires a canonical response construction, and its verdicts are defined only relative to that choice.*

Proof. Immediate from Theorem 7.9: the verdict differs between two admissible responses precisely when their cross-demands straddle θ. □

We regard Theorem 9.2 as the first experiment any implementation should perform, before the criterion is applied to biological interpretation. It is cheap: it requires one solved network and two perturbations, and it can be executed on published metabolic reconstructions [Kanehisa and Goto, 2000, Jassal et al., 2020, Orth et al., 2010] with standard tooling [Harris et al., 2020, Virtanen et al., 2020, Hagberg et al., 2008]. We have run it, and report the outcome in Section 10.1.

10 Computational validation

We implemented the framework and checked every numbered result. Minimum cuts are computed *exactly*, by enumeration over all separating vertex subsets, so that no agreement can be an artefact of an approximate solver; graphs are correspondingly small. All randomness derives from a single seeded generator, so the suite is reproducible bit-for-bit.

Two aspects of the suite deserve comment.

Table 1: Validation suite. Each row checks one numbered result. Checks are individual assertions, not trials; a trial typically contributes several. Total: 14,073 checks, all passing, in 10 s.

Result	Content	Checks
Theorem 3.4, Theorem 3.5	positive floor; no sharp identification	4,404
Theorem 3.7, Theorem 3.10	monotone local floor; conditional invariance	2,162
Theorem 4.2, Theorem 4.4	cut invariance under relabelling	1,098
Theorem 4.3	minimisers are not singletons	12
Theorem 5.3	edge set fixed; weights move	1,400
Theorem 5.6	receiver-relativity is non-vacuous	201
Theorem 6.2	distance to the floor is non-negative	3,000
Theorem 6.7, Theorem 6.8	monotone relaxation; step bound	801
Theorem 6.10, Theorem 6.11	blind certificate at positive residual	150
Theorem 7.4, Theorem 7.6	the two conflated classes	18
Theorem 5.1 (real pathways)	solved contact graphs well formed	77
Theorem 4.2 (real pathways)	invariance on solved networks	740
Theorem 7.8 (real pathways)	response-independence measurement	1
Theorem 7.2 (real pathways)	four-column verdicts well defined	9

Table 2: Response-independence measurement (Theorem 9.2) over four solved networks. Discrepancy is $|d_{\Delta\to\mathsf{B}} - d_{\Delta'\to\mathsf{B}}|$ for two admissible responses to the same species.

Quantity	Value
comparisons	35
decision threshold θ	0.01
maximum discrepancy	0.3066
mean discrepancy	0.0700
median discrepancy	0.0579
fraction within θ	0.286
Theorem 7.8 at $\theta = 0.01$	**violated**

Negative controls. Where a result has a hypothesis, we check that the hypothesis is load-bearing rather than decorative. For Theorem 3.10 we exhibit the dissolving construction of Theorem 3.8—edges of weight $1/k$ added at v—and confirm the local floor decays, so the unconditional invariance claim is refuted and the floor-stabilising hypothesis is doing real work. For Theorem 6.8 we confirm that an update which fails Theorem 6.6 (one that idles, returning D unchanged) is *rejected* rather than silently accepted, so the termination bound cannot be satisfied vacuously.

A corrected proof. The suite falsified our original witness for Theorem 7.4. The construction joined u_1, u_2 by an edge of weight β and assumed this placed them at the floor; the exact min-cut computation showed the central demand was positive, because the minimum cut ignores that edge and isolates u_1 elsewhere. The theorem is true and the pendant construction now in the proof is a correct witness. We record the episode in Theorem 7.5 because the error is precisely the one that Theorem 4.3—identity is a cut, not an edge—exists to warn against.

10.1 The response-independence measurement

We ran Theorem 9.2 on four solved metabolic and signalling networks (glycolysis, the TCA cycle, oxidative phosphorylation, and EGFR/MAPK; 35 species, 35 contact edges in total). For each species with at least two incident non-medium edges, we formed two admissible responses by raising one or the other incident edge by the same magnitude, and compared the resulting cross-demands against a fixed correspondence.

Remark 10.1 (The assumption fails on these networks). The measurement is unambiguous and negative: only 28.6% of comparisons fall within the decision threshold, and the largest discrepancy (0.307) exceeds θ by more than an order of magnitude. By Theorem 7.9, the four-column verdict on these networks is therefore *not* a function of $(x_1, \mathcal{N}_1, x_2, \mathcal{N}_2)$ alone: it depends on which admissible response was constructed.

We draw the consequence explicitly rather than softening it. As stated, Theorem 7.3 does not yield a well-defined criterion on real biochemical networks. To obtain one, the response must be *canonicalised*—the construction of $\Delta_x^{\mathcal{N}}$ must be fixed by a rule that is part of the specification, not left to the implementer. Natural candidates are the flux-weighted perturbation (raise each incident edge in proportion to its solved flux) and the uniform perturbation (raise all incident edges equally); both are determined by the solved circuit and neither introduces a free parameter. We have not evaluated whether either restores independence, and we make no claim that they do.

This is a substantive limitation of the framework as presented, and it is the reason Section 1.4 declines to claim the criterion is ready for biological interpretation. It does not affect any result in Sections 2 to 6, all of which are validated independently of Theorem 7.8.

11 Limitations

We enumerate the framework's dependencies, in decreasing order of severity.

L1. Response independence fails on the networks tested (Theorem 7.8, Theorem 10.1). We ran the deciding experiment: on four solved pathways only 28.6% of comparisons fell within the decision threshold. The four-column criterion is therefore *not* well defined on these networks without a canonicalised response rule. This is the framework's principal open problem, and it is a measured negative result rather than an untested assumption.

L2. Empirical burden is relocated, not removed. The construction requires a solved network, hence kinetic and thermodynamic parameters. Where those are wrong, the contact graph is wrong and every downstream verdict is wrong. The gain is that parameters transfer across units, whereas assay results do not; the cost is that the framework is no better than the reconstruction it runs on.

L3. Floor-stabilisation is a hypothesis (Theorem 3.9, Theorem 3.11). If a species' local floor decays to zero as the catalogue grows, it dissolves and has no intrinsic floor. The framework reports this correctly but cannot prevent it.

L4. Effective update is imposed, not derived (Theorem 6.6). Termination in Theorem 6.8 follows from it. An implementation whose update rule can stall without being at a fixed point falls outside the theorem, and would require a separate argument.

L5. Verdicts are context-scoped. By Theorem 5.6 a verdict holds relative to the networks supplied. There is no context-free annotation in this framework, by construction. This is a narrower deliverable than the field's usual ambition.

L6. Anchor selection is parameterised. The count r in Theorem 6.3 and the threshold θ are choices. We have not characterised the sensitivity of verdicts to them.

L7. No biological ground truth. Every theorem is validated computationally (Section 10) and the contact graphs of Section 10.1 are built from solved networks, but no verdict is compared against curated functional annotation. Predictions P1–P4 of Section 9 remain untested.

12 Relation to existing approaches

Alignment and homology search. Dynamic-programming alignment [Needleman and Wunsch, 1970, Smith and Waterman, 1981] and its heuristic descendants [Altschul et al., 1990, 1997] compare content. They are optimal for the question they pose and we do not improve on them. The present criterion poses a different question and is not a substitute: in particular it cannot report percent identity, E-values, or an alignment.

Profile and family models. Profile HMMs [Eddy, 1998] and substitution matrices [Henikoff and Henikoff, 1992] capture position-specific tolerance learned from families, which is an empirical proxy for the response structure we compute directly. Where family data are abundant the proxy is likely superior; where they are absent—novel units, engineered sequences, non-homologous analogues—it is unavailable and the present criterion is not.

Network and systems approaches. Flux balance and thermodynamic network analysis [Orth et al., 2010, Beard et al., 2002, Qian and Beard, 2005] supply exactly the solved networks Theorem 5.1 requires, and network biology has long held that function is a systems property [Barabási and Oltvai, 2004]. Our contribution is not that claim but a comparison criterion that follows from it and that never evaluates function.

Interpretive frameworks in clinical genetics. ACMG/AMP criteria [Richards et al., 2015] and curated resources [Landrum et al., 2018, Rehm et al., 2015] aggregate heterogeneous evidence into a categorical call. The decline outcome of Theorem 6.8(ii) is a formal counterpart of the "uncertain significance" category, with the difference that here it is a derived verdict rather than a residual bucket.

Philosophical antecedents. That reference is fixed by relation to an environment rather than by internal content is a familiar position [Quine, 1960, Putnam, 1975]. We use it only as orientation; no argument here depends on it.

13 Conclusion

Two strands of a duplex read as different codon sequences and carry one message. That fact, which requires no theory, entails that the message is not a property of a codon string. We have taken the entailment seriously.

Formalising individuation in finite weighted graphs, we derived a strictly positive floor on the cost of telling any part from the rest (Theorem 3.4), showed that the resulting identity is a minimum cut that is invariant under relabelling and never a point (Theorems 4.2 and 4.3), and concluded that labels—sequences among them—carry no invariant (Theorem 4.4). We then showed that function is not invariant either: it is receiver-relative, no single value is consistent across contexts (Theorem 5.6), and attributing one to a unit presupposes a completability that biological context does not have (Theorem 5.8).

What remains invariant is the correspondence between two systems, and we gave a procedure that certifies it without evaluating function. The relaxation is monotone (Theorem 6.7), terminates or declines with no third outcome (Theorem 6.8), and its fixed point certifies correspondence blind (Theorem 6.10) at positive residual (Theorem 6.11). Adding response columns yields a criterion (Theorem 7.3) that separates the two classes sequence comparison provably conflates: aligned content with divergent effect (Theorem 7.4) and divergent content with aligned effect (Theorem 7.6).

The framework's deliverable is narrower than annotation as usually conceived: verdicts are scoped to a supplied network, and one of four outcomes is an explicit refusal. We regard both as features. A criterion that returns a number for every input cannot distinguish a confident call from an undecidable one, and in clinical interpretation that distinction is the one that matters.

The construction rests on one assumption—that the cross-demand is independent of which admissible response is constructed—and we ran the experiment that decides it. On four solved metabolic and signalling networks the assumption *fails*: only 28.6% of comparisons fall within the decision threshold (Table 2). We therefore do not claim a working criterion for biological interpretation. What we claim is a correct account of why sequence cannot carry function, a construction that would certify functional correspondence blind if the response were canonical, and a precise identification of the one thing that must be supplied to make it so.

We think reporting this is more useful than withholding it. The negative result localises the remaining work exactly: it is not that the framework needs better parameters or more data, but that the map from a unit to the perturbation it induces must be specified by a rule rather than chosen. That is a well-posed problem, and Theorem 10.1 names two candidate rules for it.

Author contributions

Conception, formal development, and writing: K.F.S.

Conflicts of interest

None declared.

Data and code availability

The validation suite of Section 10 accompanies this paper in `validation/`, with the solved contact maps it consumes in `data/`. Running `validation/run_all.py` regenerates every number reported here, including Table 2, deterministically from the seed recorded in the source.

References

Saul B. Needleman and Christian D. Wunsch. A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of Molecular Biology*, 48(3): 443–453, 1970. doi: 10.1016/0022-2836(70)90057-4.

Temple F. Smith and Michael S. Waterman. Identification of common molecular subsequences. *Journal of Molecular Biology*, 147(1):195–197, 1981. doi: 10.1016/0022-2836(81)90087-5.

Stephen F. Altschul, Warren Gish, Webb Miller, Eugene W. Myers, and David J. Lipman. Basic local alignment search tool. *Journal of Molecular Biology*, 215(3):403–410, 1990. doi: 10.1016/S0022-2836(05)80360-2.

Stephen F. Altschul, Thomas L. Madden, Alejandro A. Schäffer, Jinghui Zhang, Zheng Zhang, Webb Miller, and David J. Lipman. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25(17):3389–3402, 1997. doi: 10.1093/nar/25.17.3389.

Steven Henikoff and Jorja G. Henikoff. Amino acid substitution matrices from protein blocks. *Proceedings of the National Academy of Sciences*, 89(22):10915–10919, 1992. doi: 10.1073/pnas.89.22.10915.

Sean R. Eddy. Profile hidden Markov models. *Bioinformatics*, 14(9):755–763, 1998. doi: 10.1093/bioinformatics/14.9.755.

Chava Kimchi-Sarfaty, Jung Mi Oh, In-Wha Kim, Zuben E. Sauna, Anna Maria Calcagno, Suresh V. Ambudkar, and Michael M. Gottesman. A "silent" polymorphism in the MDR1 gene changes substrate specificity. *Science*, 315(5811):525–528, 2007. doi: 10.1126/science.1135308.

Zuben E. Sauna and Chava Kimchi-Sarfaty. Understanding the contribution of synonymous mutations to human disease. *Nature Reviews Genetics*, 12(10):683–691, 2011. doi: 10.1038/nrg3051.

Joshua B. Plotkin and Grzegorz Kudla. Synonymous but not the same: the causes and consequences of codon bias. *Nature Reviews Genetics*, 12(1):32–42, 2011. doi: 10.1038/nrg2899.

Michael Y. Galperin, D. Roland Walker, and Eugene V. Koonin. Analogous enzymes: independent inventions in enzyme evolution. *Genome Research*, 8(8):779–790, 1998. doi: 10.1101/gr.8.8.779.

Marina V. Omelchenko, Michael Y. Galperin, Yuri I. Wolf, and Eugene V. Koonin. Non-homologous isofunctional enzymes: a systematic analysis of alternative solutions in enzyme evolution. *Biology Direct*, 5:31, 2010. doi: 10.1186/1745-6150-5-31.

Russell F. Doolittle. Convergent evolution: the need to be explicit. *Trends in Biochemical Sciences*, 19(1):15–18, 1994. doi: 10.1016/0968-0004(94)90167-8.

James D. Watson and Francis H. C. Crick. Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid. *Nature*, 171(4356):737–738, 1953. doi: 10.1038/171737a0.

Erwin Chargaff. Chemical specificity of nucleic acids and mechanism of their enzymatic degradation. *Experientia*, 6(6):201–209, 1950. doi: 10.1007/BF02173653.

Reinhard Diestel. *Graph Theory*. Springer, 5 edition, 2017. doi: 10.1007/978-3-662-53622-3.

L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi: 10.4153/CJM-1956-045-5.

Mechthild Stoer and Frank Wagner. A simple min-cut algorithm. *Journal of the ACM*, 44(4):585–591, 1997. doi: 10.1145/263867.263872.

R. E. Gomory and T. C. Hu. Multi-terminal network flows. *Journal of the Society for Industrial and Applied Mathematics*, 9(4):551–570, 1961. doi: 10.1137/0109047.

Daniel A. Beard, Shou-dan Liang, and Hong Qian. Energy balance for analysis of complex metabolic networks. *Biophysical Journal*, 83(1):79–86, 2002. doi: 10.1016/S0006-3495(02)75150-3.

Hong Qian and Daniel A. Beard. Thermodynamics of stoichiometric biochemical networks in living systems far from equilibrium. *Biophysical Chemistry*, 114(2-3):213–220, 2005. doi: 10.1016/j.bpc.2004.12.001.

Jeffrey D. Orth, Ines Thiele, and Bernhard Ø. Palsson. What is flux balance analysis? *Nature Biotechnology*, 28(3):245–248, 2010. doi: 10.1038/nbt.1614.

Minoru Kanehisa and Susumu Goto. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1):27–30, 2000. doi: 10.1093/nar/28.1.27.

Bijay Jassal, Lisa Matthews, Guilherme Viteri, Chuqiao Gong, Pascual Lorente, Antonio Fabregat, Konstantinos Sidiropoulos, Justin Cook, Marc Gillespie, Robin Haw, et al. The reactome pathway knowledgebase. *Nucleic Acids Research*, 48(D1):D498–D503, 2020. doi: 10.1093/nar/gkz1031.

Ida Schomburg, Antje Chang, and Dietmar Schomburg. BRENDA, enzyme data and metabolic information. *Nucleic Acids Research*, 30(1):47–49, 2002. doi: 10.1093/nar/30.1.47.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

Charles H. Bennett. The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982. doi: 10.1007/BF02084158.

Juan M. R. Parrondo, Jordan M. Horowitz, and Takahiro Sagawa. Thermodynamics of information. *Nature Physics*, 11(2):131–139, 2015. doi: 10.1038/nphys3230.

GTEx Consortium. The GTEx consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509):1318–1330, 2020. doi: 10.1126/science.aaz1776.

David N. Cooper, Michael Krawczak, Cynthia Förster-Gibson, and Stylianos E. Antonarakis. Where genotype is not predictive of phenotype: towards an understanding of the molecular basis of reduced penetrance in human inherited disease. *Human Genetics*, 132(10):1077–1130, 2013. doi: 10.1007/s00439-013-1331-2.

Ben Lehner. Molecular mechanisms of epistasis within and between genes. *Trends in Genetics*, 27(8):323–331, 2011. doi: 10.1016/j.tig.2011.05.007.

Julià Domingo, Pablo Baeza-Centurion, and Ben Lehner. The causes and consequences of genetic interactions (epistasis). *Annual Review of Genomics and Human Genetics*, 20:433–460, 2019. doi: 10.1146/annurev-genom-083118-014857.

Michael Costanzo, Benjamin VanderSluis, Elizabeth N. Koch, Anastasia Baryshnikova, Carles Pons, Guihong Tan, Wen Wang, Matej Usaj, Julia Hanchard, Susan D. Lee, et al. A global genetic interaction network maps a wiring diagram of cellular function. *Science*, 353(6306), 2016. doi: 10.1126/science.aaf1420.

Teri A. Manolio, Francis S. Collins, Nancy J. Cox, David B. Goldstein, Lucia A. Hindorff, David J. Hunter, Mark I. McCarthy, Erin M. Ramos, Lon R. Cardon, Aravinda Chakravarti, et al. Finding the missing heritability of complex diseases. *Nature*, 461(7265):747–753, 2009. doi: 10.1038/nature08494.

Evan A. Boyle, Yang I. Li, and Jonathan K. Pritchard. An expanded view of complex traits: from polygenic to omnigenic. *Cell*, 169(7):1177–1186, 2017. doi: 10.1016/j.cell.2017.05.038.

Sue Richards, Nazneen Aziz, Sherri Bale, David Bick, Soma Das, Julie Gastier-Foster, Wayne W. Grody, Madhuri Hegde, Elaine Lyon, Elaine Spector, Karl Voelkerding, and Heidi L. Rehm. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*, 17(5):405–424, 2015. doi: 10.1038/gim.2015.30.

Melissa J. Landrum, Jennifer M. Lee, Mark Benson, Garth R. Brown, Chen Chao, Shanmuga Chitipiralla, Baoshan Gu, Jennifer Hart, Douglas Hoffman, Wonhee Jang, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

Lily Hoffman-Andrews. The known unknown: the challenges of genetic variants of uncertain significance in clinical practice. *Journal of Law and the Biosciences*, 4(3):648–657, 2017. doi: 10.1093/jlb/lsx038.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.

Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, 2008.

Albert-László Barabási and Zoltán N. Oltvai. Network biology: understanding the cell's functional organization. *Nature Reviews Genetics*, 5(2):101–113, 2004. doi: 10.1038/nrg1272.

Heidi L. Rehm, Jonathan S. Berg, Lisa D. Brooks, Carlos D. Bustamante, James P. Evans, Melissa J. Landrum, David H. Ledbetter, Donna R. Maglott, Christa L. Martin, Robert L. Nussbaum, et al. ClinGen—the clinical genome resource. *New England Journal of Medicine*, 372(23):2235–2242, 2015. doi: 10.1056/NEJMsr1406261.

Willard Van Orman Quine. *Word and Object*. MIT Press, 1960.

Hilary Putnam. The meaning of 'meaning'. *Minnesota Studies in the Philosophy of Science*, 7: 131–193, 1975.